

Figure 19.5 Microbial diversity from sandstone and granite from the McMurdo Dry Valleys, Antarctica

DO YOU KNOW?

NASA is developing the Mars 2020 astrobiology to investigate an astrobiologically relevant ancient environment on Mars, its surface geological processes and the possibility of past life on Mars and preservation of biosignatures within accessible geological materials.

Points to Remember

- Lamarck proposed that the acquired characters are passed on to the offsprings in the next generation
- Internal vital force, environment and new needs, use and disuse theory and inheritance of acquired characters are the main principles of Lamarckism.
- Overproduction, struggle for existence, variations, survival of the fittest or Natural selection and origin of species are the main postulates of Darwinism.
- Each species tends to produce large number of offsprings, but only the fittest can survive.
- Homologous, analogous organs and embryological evidences explain evolutionary relationships.
- Some traits in organisms would be similar because they are inherited from a common ancestor.
- Fossils are evidences of ancient life forms or ancient habitats which have been preserved by natural processes.
- Ethnobotanical importance of various types of plants are know through traditional knowledge.
- Astrobiology/exobiology is the science which looks for the presence of extra terrestrial life in the universe

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TEXTBOOK EVALUATION

I Choose the correct answer

1. Biogenetic law states that _____
   a. Ontogeny and phylogeny go together
   b. Ontogeny recapitulates phylogeny
   c. Phylogeny recapitulates ontogeny
   d. There is no relationship between phylogeny and ontogeny

2. The 'use and disuse theory' was proposed by _____.
   a. Charles Darwin
   b. Ernst Haeckel
   c. Jean Baptiste Lamarck
   d. Gregor Mendel

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